

FILIPPO MONTI

Los Angeles, CA, USA | <https://fil-monti.github.io>

Emails: fmonti@ucla.edu (work) | monti.filippo97@gmail.com (personal)

Research Interests: *Bayesian statistics, computational statistics, nonparametric methods, phylogenetics.*

Education

University of California, Los Angeles (UCLA), Los Angeles, CA, USA

Ph.D. Candidate in Biostatistics

2022 – Fall 2026 (expected)

Advisor: Prof. Marc A. Suchard

Bocconi University, Milan, Italy

M.Sc. in Economic and Social Sciences (ESS), *110/110 cum laude*

2019 – 2022

Thesis: *Bayesian nonparametric methods in Biostatistics*

Supervisor: Prof. Igor Prünster

Bocconi University, Milan, Italy

B.Sc. in Economic and Social Sciences (CLES), *110/110 cum laude*

2016 – 2019

Awards and Fellowships

Graduate Division Fellowship (funded), UCLA

2022 – 2026

Graduate Summer Research Mentorship Program (funded), UCLA

Summer 2023

Graduate Merit Award (full tuition waiver), Bocconi University

2019 – 2021

UniCredit Foundation Grant (full tuition waiver), BSE Summer School

Summer 2021

Publications and Working Papers

Frumence, E., Klitting, R., Serres, K., Shao, Y., **Monti, F.**, Vincent, M., Gill, M. S., Suchard, M. A., Lemey, P., de Lamballerie, X., Jaffar-Bandjee, M.-C., and Dellicour, S. (2026). *Unravelling the epidemiological and dispersal dynamics of the 2024–2025 chikungunya virus outbreak on Réunion island*. In press, *Proceedings of the National Academy of Sciences*.

Monti, F., Ji, X., Shao, Y., and Suchard, M. A. (2026). *Nonparametric Modeling of Continuous-Time Markov Chains with Scalable Exact and Approximate Gradients*. R&R at *JRSS-B*.

Ji, X., Redelings, B., Su, S., Bao, H., Deng, W.-M., **Monti, F.**, Hong, S. L., Baele, G., Lemey, P., and Suchard, M. A. (2026). *Detecting Evolutionary Change-Points with Branch-Specific Substitution Models and Shrinkage Priors*. Minor revision at *Systematic Biology*.

Monti, F., Faria, N. R., Ji, X., Lemey, P., Kraemer, M. U. G., and Suchard, M. A. (2026). *Non-Linear Drivers of Population Dynamics: a Nonparametric Coalescent Approach*. Under review.

Monti, F., Holbrook, A., Glatt-Holtz, N. E., and Suchard, M. A. (2026). *Stable Matrix Parametrizations and Structured Adjoints for Ornstein–Uhlenbeck Processes*. Under review.

Monti, F., Lemey, P., and Suchard, M. A. *Markov Modulation via Reward Augmentation*. Manuscript in preparation.

Monti, F., Lemey, P., and Suchard, M. A. *Investigating the drivers of time-varying rates of evolution*. Manuscript in preparation; invited paper for the *Methods in Ecology and Evolution* Special Feature: “Time-dependency in micro- and macroevolutionary rates.”

Invited Talks and Conference Presentations

CGSI 2026 - Computational Genomics Summer Institute, University of California Los Angeles, Los Angeles, CA, USA. Talk: *Scalable Inference for Continuous-Time Markov Processes*.

SysBio 2025 – Annual Systems Biology of Infectious Diseases Meeting, Rockville, MD, USA. Invited talk: *Flexibly Modeling Viral Substitution and Coalescent Processes with Covariates*.

BISP 2025 – Bayesian Inference in Stochastic Processes, Milan, Italy. Poster: *Nonparametric Modeling of Continuous-Time Markov Chains*.

ISBA World Meeting 2024, Venice, Italy. Poster: *Nonparametric Modeling of Continuous-Time Markov Chains*.

Teaching

Applied Bayesian Statistics School in Como (Italy)

Lecturer (jointly with Prof. Suchard)

Bayesian phylogenetics and infectious diseases

Summer 2024

University of California, Los Angeles (UCLA)

Teaching Assistant

Biostat 250C: Multivariate Biostatistics (Ph.D. level)

Apr 2026 – Jun 2026

Biostat 250B: Linear Models (Ph.D. level)

Jan 2026 – Mar 2026

Biostat 250A: Linear Models (Ph.D. level)

Sep 2025 – Dec 2025

Biostat 100A/B: Introduction to Biostatistics

Sep 2022 – Sep 2023

Other Appointments

University of California, Los Angeles (UCLA), Los Angeles, CA, USA

Graduate Research Assistant, Suchard Group

Jul 2022 – Present

Bocconi University, Milan, Italy

Fellow, Bayesian Learning Laboratory (BayesLab)

2021 – 2022

<i>Visiting Student</i> , BIDS-IGIER	2020 – 2021
<i>Research Intern</i> , IGIER (Supervisor: Prof. Marinacci)	Summer 2018
Harvard University , Cambridge, MA, USA	
<i>Fellow</i> , Social Economics Lab	2020 – 2021
<i>Research Assistant</i> (hosted at IGIER, Bocconi)	Nov 2020 – Jul 2021
European Commission , Luxembourg	
<i>Intern</i> , DG ECFIN, Division L3: Risk Management	Jun 2019 – Aug 2019

Software

BEAST	2024 – Present
Contributor to BEAST, developing scalable likelihood and gradient-based methods for Bayesian phylogenetic inference.	

Service and Leadership

ISBA Bayesian Nonparametrics (BNP 14) , Los Angeles, CA, USA	
<i>Volunteer</i> , organizational support	Jul 2025
Bocconi University , Milan, Italy	
President, Economic Society for Bocconi Students	Sep 2020 – Sep 2021
Student Representative, Council of the Department of Finance	2017 – 2019
Student Representative, Program Committee	2017 – 2018

Other Activities

Agesci (Italian Scout Association), Forlì, Italy	2006 – 2018
Volunteer, education for disadvantaged children, Baia Mare, Romania	Aug 2016
Volunteer, elderly care home, Forlì, Italy	Sep 2015 – Apr 2016

Last updated: August 19, 2026